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A Scalable and Lightweight Authentication Architecture for the Internet of Things (IoT) in Smart Home Applications

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***Abstract*— In the evolving landscape of smart home applications, ensuring robust and efficient authentication within the Internet of Things (IoT) environments is paramount. This paper presents a scalable, lightweight authentication architecture for IoT-driven smart homes. The proposed architecture, drawing inspiration from cloud computing mechanisms, introduces a trusted, smart home agent, integrating the Data-Information-Knowledge-Wisdom (DIKW) model to facilitate efficient and secure data exchange within smart homes. Our solution addresses the challenges of minimal computation and processing overheads, essential in resource-constrained IoT devices, and focuses on device-level authentication to reduce computational costs. A comparative analysis with existing models demonstrates the superiority of our architecture in terms of computational efficiency, making it an ideal candidate for scalable deployments. The architecture successfully navigates the complexities of ensuring secure communication among autonomous IoT devices, balancing the dual needs of computational efficiency and robust security, thereby promising resilience against common security threats in the smart home ecosystem.**

Keywords*— Internet of Things, Smart Home, Authentication Architecture, Lightweight Authentication, DIKW Model, Security, IoT Devices.

I. INTRODUCTION

The advent of the Internet of Things (IoT) has significantly impacted how we live, particularly in smart home applications. These smart home environments, enriched with interconnected IoT devices, have revolutionised daily living by enhancing efficiency, convenience, and overall quality of life [1]. However, the proliferation of IoT devices in such settings introduces complex challenges, notably securing communication and ensuring data privacy. Authentication, the process of verifying the identity of a device or user, emerges as a critical concern in this context. In smart home environments, devices frequently operate autonomously, and securing these interactions is paramount [2], [3].

Current research in IoT security explores various authentication models. These models range from ID-based schemes, predominantly used in smart city applications [4], to mutual authentication strategies, which have seen application in health sectors and cloud computing environments [5], [6]. Despite their effectiveness in specific contexts, these strategies often fall short in smart home scenarios, where the need for minimal computational intensity is paramount due to the limited resources of IoT devices [7], [8].

This paper proposes a scalable and lightweight authentication architecture explicitly designed for IoT-driven smart homes to address these challenges. The architecture aims to balance computational efficiency and robust security, a critical requirement in smart home IoT networks. It introduces a novel approach incorporating a trusted, smart home agent, leveraging the Data-Information-Knowledge-Wisdom (DIKW) model to facilitate a secure and efficient data exchange [9].

This paper is structured to understand the proposed architecture and its benefits comprehensively. Following this introduction, Section II discusses related work to contextualise our contribution within the broader IoT security landscape. Section III outlines the methodology employed in the broader research context of this work. Section IV details our proposed lightweight authentication architecture, describing its components and operational mechanisms. Section V presents a comparative analysis of our architecture against existing solutions, highlighting its advantages in terms of performance and efficiency. Finally, Section VI concludes the paper by summarising the essential findings and suggesting avenues for future research in this domain.

II. RELATED WORK

The landscape of IoT in smart home applications has been a fertile ground for research, particularly in authentication and security. This section reviews relevant literature, highlighting key contributions and identifying gaps our research seeks to address.

A. IoT Authentication Models

The importance of robust authentication models in IoT settings has been widely acknowledged in the literature. Chauhan et al. [10] underscored the need for authentication mechanisms tailored to the resource-constrained nature of IoT devices, using Recurrent Neural Networks for breathing-based authentication as a case study.

B. Mutual Authentication Strategies

Mutual authentication, where both communicating parties authenticate each other, has been explored extensively. The broader challenges of IoT device authentication, which include mutual authentication aspects in healthcare and different contexts, have been documented as a growing concern, with vulnerabilities and threats needing urgent attention[11].

C. Lightweight Authentication Solutions

The concept of lightweight authentication has emerged as a response to the computational limitations of traditional methods. A survey of smart home device security, which includes discussions on lightweight authentication, suggests that while there have been developments, there remains a lack of focus on smart home applications precisely [12].

D. The Gap in Existing Research

While existing research has made significant strides in IoT authentication, there remains a notable gap in developing solutions tailored explicitly for smart home environments. The unique challenges of these settings, such as the need for low computational overhead and high scalability, are yet to be fully addressed. Design considerations for usable authentication in smart homes have been discussed, reflecting on implications and suggesting directions for future research [13].

E. Our Contribution

Building on the foundations laid by previous research, our paper introduces a scalable and lightweight authentication architecture for IoT-driven smart homes. We propose integrating a trusted, smart home agent and leveraging the DIKW model to overcome the limitations of existing models. This architecture is designed to offer an efficient and secure solution for smart homes' dynamic and resource-constrained context.

III. Methodology

The development of a lightweight authentication architecture for IoT-driven smart homes was approached through a blend of theoretical research and practical application. This methodology was designed to methodically tackle the intricacies of IoT security challenges, with a concentrated effort on authentication.

A. Conceptual Framework and Theoretical Modelling

The initial phase of the research involved establishing a conceptual framework, identifying core challenges in IoT security, and focusing specifically on authentication issues. The Data-Information-Knowledge-Wisdom (DIKW) model underpins the architectural design, with the Smart Home Agent (SHA) posited as a central mediator for authentication processes. This framework shaped the scope and directed the research rationale.

B. Smart Home Environment Analysis and Modelling

An in-depth analysis of smart home environments was executed [14], entailing a digital simulation to emulate the interaction between IoT devices and the SHA. This step tested the efficiency of the authentication processes. It informed the creation of security-aware specifications for smart homes by identifying potential bottlenecks and modelling smart home requirements, applications, and future projections.

C. Threat Landscape and Architecture Review

The research advanced with identifying and categorising threats specific to smart home IoT applications [15], creating a security threat taxonomy. Concurrently, a review of existing IoT authentication architectures was conducted to assess their suitability and inform the design of a tailored, cost-effective solution for smart homes [16].

D. Design and Development of Authentication Architecture

Employing Design Science Research and Formal Methods, the proposed authentication architecture was rigorously developed through iterative design. This involved refining the SHA's role to alleviate computational burdens from IoT devices, catering to unsupervised operations, and focusing on the practicalities of real-world implementation.

E. Security Verification and Performance Evaluation

The security features of the architecture were evaluated using the SCYTHER tool for formal verification and testing against prevalent threats such as replay, man-in-the-middle, and impersonation attacks. A comparative analysis measured the architecture's performance, precisely its computational efficiency and scalability against established models.

F. Practical Implementation and Testing

Following theoretical validation, the architecture was implemented within a test network of IoT devices to observe its real-world performance. This phase encompassed device enrolment, authentication processes, and data exchange, ensuring the architecture's functionality met the anticipated security standards.

The comprehensive methodology led to developing an authentication architecture that is secure and efficient and exhibits significant scalability. It ensures robust protection against a broad spectrum of security threats while maintaining minimal computational overhead, thus reflecting a successful application of the methodology as substantiated by the Performance Analysis.

IV. Proposed Authentication Architecture

To mitigate computational strain on Internet of Things (IoT) devices within smart home networks, we introduce an innovative architecture centred around a trusted entity, the Smart Home Agent (SHA). This SHA acts as an intermediary, facilitating seamless mutual authentication between devices and virtual storage, thus ensuring secure communication.

A. Roles within the Architecture

The architecture is predicated on three core roles:

1. **Virtual Storage (VS):** As the repository, the VS provides services to the IoT devices ('things') and maintains security credentials.
2. **The 'Things'**: Individual IoT devices within the network, ranging from 'Thing 1' to 'Thing n', each requiring authentication to join and operate within the home network.
3. **Smart Home Agent (SHA):** Operating as the keystone of the architecture, the SHA is a trusted security layer that generates security parameters, distributes identities (IDs) to authorised users and VS, and mediates all authentication processes.

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B. Operational Phases

The architecture operates in two main phases:

1) Bootstrapping Phase

During the initial request phase, the SHA generates random numbers for each 'thing', and the VS is denoted as Nv and N1-n, respectively.

A. The request phase (represented as (1) in Fig. 1)

[1]. Step 1: SHA generates a random number for the thing requesting to be added to the network and another one for the virtual storage, hence N_v (for the virtual server) and $N_{1\text{-}n}$ (for the thing).

[2]. Step 2: The thing and the virtual storage generate identity requests $ID_{r,v}$ and $ID_{r,\,1\text{-}n}$ respectively.

[3]. Step 3: The thing and the virtual storage server send {IDr,1-n, $N_{1\text{-}n}$} and {IDr,v, Nv} to the SHA, respectively.

B. The reply phase (represented as (2) in Fig.1)

[1]. Step 1: According to the received request, SHA will check the request's legitimacy and generate an ID or discard it.

[2]. Step 2: If the request passes the verification, SHA will generate the identity $ID_{1\text{-}n}$ and ID_v, a security parameter k and a timestamp $TS_{1\text{-}n}$ and TS_v for the thing and the virtual storage, respectively.

[3]. Step 3: To ensure the possibility of verifying the communication during the authentication phase as well as decreasing the computation on the thing side and the virtual storage side, SHA will compute $A = h(ID_{1\text{-}n} \| k) \oplus N_{1\text{-}n}$ and $B = h(ID_v \| k) \oplus N_v$ and then sends $\{ID_{1\text{-}n}, B, k, TS_{1\text{-}n}\}$ and $\{ID_v, A, k, TS_v\}$ to the thing and the Virtual storage respectively.

Subsequent identity requests from the 'things' and the VS are then processed by the SHA, which either generates IDs for valid requests or discards those not meeting the legitimacy criteria.

2) Authentication Phase

This phase includes the Thing Authentication and the Virtual Storage Authentication processes.

The authentication phase witnesses two primary operations as follows: -

A. Thing Authentication Phase (represented as (3) in Fig. 1)

[1]. Step 1: The thing sends $\{ID_{1\text{-}n}, TS_{1\text{-}n}, N_{1\text{-}n}\}$ to SHA.

[2]. Step 2: After receiving the parameter combination from SHA, the Virtual storage will check if the $TS_{1\text{-}n}$ is valid, then compute $p_{1\text{-}n} = A \oplus N_{1\text{-}n}$ and $p_{1\text{-}n} = h(ID_{1\text{-}n} \| k)$.

[3]. Step 3: if $p_{1\text{-}n} = p_{1\text{-}n}$, the thing is confirmed to pass the authentication; otherwise, the Virtual server ensures this thing is flagged as illegal and denies it access to its requests.

B. Virtual Storage Authentication Phase (represented as (3) in Fig. 1)

[1]. Step 1: The Virtual storage sends $\{ID_v, TS_v, N_v\}$ to SHA.

[2]. Step 2: After receiving the parameter combinations, SHA will check whether the TS_v is valid. If TS_v is valid, the thing will compute $p_v = B \oplus N_v$ and $P_v = h(ID_v \| k)$.

[3]. Step 3: If $p_v = p_v$, the Virtual storage is confirmed to have passed the authentication.

The SHA plays a pivotal role in reducing computational demands by pre-processing authentication parameters, thus streamlining the overall security protocol.

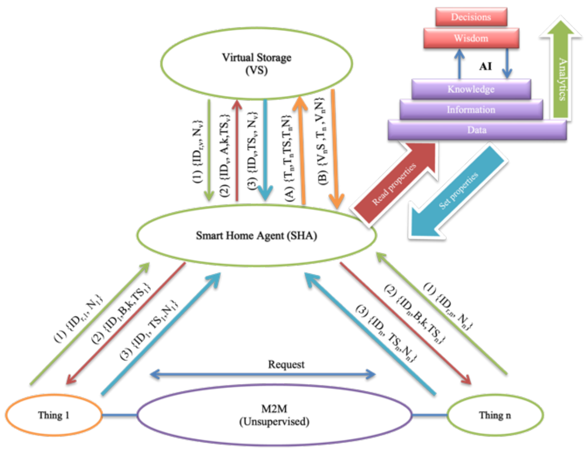

Fig. 1. Proposed Authentication Architecture (AA)

C. Notations and Interpretations

We have delineated the notation used within our architecture to understand the authentication process.

TABLE I. THE NOTATIONS USED FOR AA

Notation	Interpretation
ID_v	Virtual storage identity
$ID_{1\text{-}n}$	Thing's identity from 1 to n
$ID_{r,v}$	Virtual server's identity request
N_v	A random number chosen by the virtual storage
$N_{1\text{-}n}$	A random number chosen by thing 1 to n
K	A security parameter chosen by the smart home agent
$\oplus$	XOR operation
h(-)	A secure one-way hash function
$\|$	String concatenation operation
TS_v	Timestamp of the virtual storage
$TS_{1\text{-}n}$	Timestamp of respective thing from 1 to n
TS_a	Timestamp of smart home agent

Through this structured approach, our architecture ensures a robust yet lightweight authentication mechanism, pivotal for the integrity and security of IoT-driven smart home environments.

V. PERFORMANCE ANALYSIS

To validate the efficiency of our proposed authentication scheme, we conducted a rigorous comparative analysis against established authentication models. Our study focused on

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quantifying the computational overhead on IoT devices ('things') and the virtual server, which is a critical metric for performance in resource constrained IoT environments.

A. Computational Overhead on IoT Devices

Our proposed architecture eliminates the need for computationally intensive operations on the IoT device side. This is a significant enhancement over existing schemes, as demonstrated in the following comparison table:

TABLE II. COMPUTATIONAL OVERHEAD COMPARISON FOR IOT DEVICES

Operation	Schemes			
	Yang et al [17]	*Yang et al [18]*	*Shen et al [19]*	***Proposed Scheme***
XOR (x)	4	2	1	**0**
Hash Function (y)	6	4	1	**0**
Concatenation function (z)	4	6	1	**0**
Total Computation Cost	4x+6y+4z	2x+6y+6z	x+y+z	**0x+0y+0z**

The results underscore the absence of XOR, hash, and concatenation operations in our scheme, leading to a computation cost of zero on the IoT device side, thereby substantiating its lightweight nature.

B. Computational Overhead on Virtual Server

Similarly, on the virtual server side, our scheme shows an improved computation cost profile that aligns with the most efficient of the established methods:

TABLE III. COMPUTATIONAL OVERHEAD COMPARISON FOR VIRTUAL SERVER

Operation	Schemes			
	Yang et al [17]	*Yang et al [18]*	*Shen et al [19]*	***Proposed Scheme***
XOR (x)	4	2	1	**1**
Hash Function (y)	6	4	1	**1**
Concatenation function (z)	4	6	1	**1**
Total Computation Cost	4x+6y+4z	2x+6y+6z	x+y+z	**x+y+z**

The computation cost for the virtual server within our proposed scheme matches that of the most efficient existing model [19], which highlights the enhanced performance of our architecture in terms of computational demands.

C. Analysis and Discussion

The comparative analysis distinctly illustrates the reduced computational load of our proposed scheme, a paramount advantage for IoT environments where computational resources often constrain devices. By shifting the bulk of the computational work to the Smart Home Agent, we ensure that the devices and the virtual server are not burdened with heavy cryptographic operations, thus enhancing the overall system performance and scalability.

Furthermore, the zero-computation cost on the IoT devices improves performance and extends the devices' operational lifespan by minimising power consumption—a critical benefit for smart home applications.

In conclusion, our proposed authentication scheme emerges as a highly competitive solution, offering significant computational efficiency gains, thereby confirming its suitability for contemporary smart home IoT environments.

VI. CONCLUSION

In conclusion, this paper introduces a novel lightweight authentication architecture that caters to the unique demands of IoT-driven smart home environments. The proposed solution synthesises insights from existing authentication models and our innovative approach to computational efficiency and security.

Our architecture's design is particularly attentive to the computational limitations inherent to IoT devices. By allocating the bulk of computational tasks to a trusted Smart Home Agent (SHA), we effectively minimise the processing load on individual devices. This strategy enhances the network's operational efficiency and extends the lifespan of IoT devices by conserving their battery life and processing capabilities.

Regarding security, our developed architecture utilises established cryptographic operations to fortify the network against prevalent threats. It has been rigorously designed to counteract various security vulnerabilities, including replay, man-in-the-middle, and impersonation attacks. The resilience of our system is underpinned by its ability to seamlessly integrate with the dynamic and heterogeneous nature of smart home ecosystems.

Furthermore, the architecture's scalability is a testament to its suitability for future expansion within the IoT domain. As smart home technologies evolve and connect devices proliferate, our architecture can adapt and scale, accordingly, ensuring its long-term applicability and reliability.

The performance analysis underscores the proposed architecture's superiority over existing models regarding reduced computational overhead. Through empirical evidence, we demonstrate that our system significantly decreases the computational burden on IoT devices, thus positioning it as a leading lightweight solution within the smart home security landscape.

Looking forward, we anticipate that the principles and methodologies outlined in this paper will serve as a foundational framework for future research in IoT security. The architecture's flexible design provides ample opportunities for further enhancement and innovation, particularly in machine learning and artificial intelligence, which could offer predictive security measures based on user behaviour and anomaly detection.

In essence, our work contributes a significant leap forward in pursuing a secure, efficient, and intelligent smart home ecosystem, ensuring that user convenience does not come at the cost of security.

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